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Motor incorporating magnetic bearings

Abstract

An air cycle machine includes a turbine section, a compressor section, an axial flux magnetic gear system electromechanically coupling the turbine section to the compressor section, and a stabilization system for at least one shaft of the air cycle machine. The axial flux magnetic gear system includes a stator, two rotors, a winding system, and a control module. Each rotor interacts with the stator such that the rotor rotates when the axial flux magnetic gear system is exposed to an electrical current. The winding system stabilizes the position of the rotors along their rotational axes. The control module is configured to supply the electrical current to at least one of the winding system, the stator, and the rotors to drive rotation of the rotors. The stabilization system stabilizes the position of the at least one shaft of the air cycle machine along the shaft's rotational axis.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
RE32100	12/1985	Rannenberg	N/A	N/A
5490645	12/1995	Woodhouse	N/A	N/A
5572119	12/1995	Taylor	N/A	N/A
5709103	12/1997	Williams	N/A	N/A
RE36101	12/1998	Andres et al.	N/A	N/A
6058715	12/1999	Strang et al.	N/A	N/A
6608418	12/2002	Andres et al.	N/A	N/A
6735953	12/2003	Wolfe et al.	N/A	N/A
6776002	12/2003	Ho	62/401	B64D 13/08
8347648	12/2012	Nakazeki et al.	N/A	N/A
10174767	12/2018	Pal	N/A	N/A
2016/0087517	12/2015	Powell et al.	N/A	N/A
2017/0350405	12/2016	Sozer et al.	N/A	N/A
2020/0277079	12/2019	Retersdorf et al.	N/A	N/A
2022/0161932	12/2021	Sharpe	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
216269938	12/2021	CN	N/A
1471002	12/2003	EP	N/A

OTHER PUBLICATIONS

Z.Q. Zhu, et al., "A Novel Axial Flux Magnetically Geared Machine for Power Split Application", from IEEE Transactions on Industry Applications, 54 (6), pp. 5954-5966. cited by applicant
Extended European Search Report for EP Application No. 23210418.2, dated Apr. 22, 2024, 6 pages. cited by applicant
Extended European Search Report for EP Application No. 23211583.2, dated Apr. 24, 2024, 6 pages. cited by applicant
Extended European Search Report for EP Application No. 23211822.4, dated Apr. 24, 2024, 6 pages. cited by applicant
Notice of Allowance and Fee(s) Due dated Jun. 9, 2025, for corresponding U.S. Appl. No. 18/075,148, 23 pgs. cited by applicant

Background/Summary

BACKGROUND

- (1) The present invention relates generally to environmental control systems, and in particular to an air cycle machine configuration.
- (2) Environmental control systems condition a supply of air and deliver it to an enclosed section of an aircraft, such as a cabin or cockpit. An environmental control system can include an air cycle machine, which includes a compressor and at least one turbine, as well as at least one heat exchanger. The compressor and turbine(s) are typically located along a common shaft and accordingly rotate at the same speed (i.e., at the rotational speed of the common shaft). Conventionally, the rotational speed of a turbine and compressor system is established by balancing the amount of power the turbine(s) can generate with the amount of power consumed by the compressor. However, the compressor and turbine(s) have differing optimal rotational speeds at which they achieve peak efficiency. Operating the compressor and turbine(s) at a uniform speed results in a loss of performance and correspondingly increases the weight of the associated heat exchanger(s) and overall system weight. Additionally, the air cycle machine can require a booster compressor located upstream of the turbine in order to achieve the desired rotational speed in the compressor section, which adds additional weight to the system.
- (3) Air cycle machine typically use air bearings to support and stabilize the shaft coupling the turbine(s) to the compressor(s). However, air bearings require a flow of a cooling medium during operation, which is typically directed to the bearings through a separate ducting system. Additionally, the performance of air bearings can be negatively impacted by debris.

SUMMARY

- (4) An air cycle machine includes a turbine section, a compressor section, an axial flux magnetic gear system electromechanically coupling the turbine section to the compressor section, and a stabilization system for at least one shaft of the air cycle machine. The axial flux magnetic gear system includes a stator, a first rotor, a second rotor, a winding system, and a control module. The stator includes stator pole sections and is oriented about a stator axis. The first rotor includes a first plurality of magnets arranged radially about a first rotational axis aligned with the stator axis and configured to interact with the stator such that the first rotor rotates at a first rotational speed when the axial flux magnetic gear system is exposed to an electrical current. The second rotor includes a second plurality of magnets arranged radially about a second rotational axis aligned with the first rotational axis and being configured to interact with the stator such that the second rotor rotates at a second rotational speed when the axial flux magnetic gear system is exposed to the electrical current. The winding system stabilizes a position of the first rotor along the first rotational axis and a position of the second rotor along the second rotational axis. The control module is configured to supply the electrical current to at least one of the winding system, the stator, the first rotor, and the second rotor to drive rotation of the first rotor about the first rotational axis and the second rotor about the second rotational axis. The stabilization system stabilizes a position of the at least one shaft of the air cycle machine along a rotational axis of the at least one shaft.
- (5) As further discussed herein, an air cycle machine includes a turbine section, a compressor section, an electrical gearbox, and a stabilization system for at least one of the turbine shaft and the compressor shaft. The turbine section includes a plurality of turbine flow elements disposed about a turbine shaft. The compressor section includes a plurality of compressor flow elements disposed about a compressor shaft axially aligned with the turbine shaft. The electrical gearbox is located between, and electromechanically couples, the turbine shaft and the compressor shaft. The

electrical gearbox includes a gearbox housing and an axial flux magnetic gear system comprising disposed within the gearbox housing. The axial flux magnetic gear system includes a stator, a first rotor, a second rotor, a winding system, and a control module. The stator includes a plurality of stator pole sections oriented about a stator axis, and located between the first rotor and the second rotor. The first rotor includes a first plurality of magnets arranged radially about a first rotational axis aligned with the stator axis and configured to interact with the stator such that the first rotor rotates at a first rotational speed when the axial flux magnetic gear system is exposed to an electrical current. The first rotor is mechanically coupled to the turbine shaft. The second rotor includes a second plurality of magnets arranged radially about a second rotational axis aligned with the first rotational axis and being configured to interact with the stator such that the second rotor rotates at a second rotational speed when the axial flux magnetic gear system is exposed to the electrical current. The second rotor is mechanically coupled to the compressor shaft. The winding system stabilizes a position of the first rotor along the first rotational axis and a position of the second rotor along the second rotational axis. The control module is configured to supply the electrical current to at least one of the winding system, the stator, the first rotor, and the second rotor to drive rotation of the first rotor about the first rotational axis and the second rotor about the second rotational axis. The stabilization system stabilizes a position of the at least one of the turbine shaft and the compressor shaft along a rotational axis of the at least one of the turbine shaft and the compressor shaft.

(6) As also discussed herein, a method of operating an air cycle machine for an aircraft includes generating an electrical current and supplying the electrical current to an axial flux magnetic gear system within an electrical gearbox of the air cycle machine. The axial flux magnetic gear system rotates a compressor shaft of the air cycle machine. A position is adjusted for at least one of a turbine shaft and the compressor shaft of the air cycle machine along a rotational axis of the at least one of the turbine shaft and the compressor shaft using a stabilization system. The air cycle machine conditions an airflow for an aircraft cabin of the aircraft.

(7) The present summary is provided only by way of example, and not limitation. Other aspects of the present disclosure will be appreciated in view of the entirety of the present disclosure, including the entire text, claims, and accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic depiction of an environmental control system.

(2) FIG. 2 is a simplified cross-sectional view of an air cycle machine including an electrical gearbox.

(3) FIG. 3A is a schematic depiction of a high-speed rotor of a magnetic gear system.

(4) FIG. 3B is a schematic depiction of a low-speed rotor of a magnetic gear system.

(5) FIG. 3C is a schematic depiction of a stator of a magnetic gear system.

(6) FIG. 3D is a schematic depiction of the interaction between a rotor and a stator of a magnetic gear system.

(7) FIG. 4 is a schematic cross-sectional depiction of an electrical gearbox including a magnetic gear system.

(8) FIG. 5 is a schematic depiction of a bearing system for a shaft system of the electrical gearbox of FIG. 4.

(9) While the above-identified figures set forth one or more embodiments of the present disclosure, other embodiments are also contemplated, as noted in the discussion. In all cases, this disclosure presents the invention by way of representation and not limitation. It should be understood that numerous other modifications and embodiments can be devised by those skilled in the art, which

fall within the scope and spirit of the principles of the invention. The figures may not be drawn to scale, and applications and embodiments of the present invention may include features and components not specifically shown in the drawings.

DETAILED DESCRIPTION

(10) An electrical gearbox can be integrated into an air cycle machine (ACM) for an aircraft to allow a difference in rotational speeds between the turbine and compressor sections. The control module for the electrical gearbox can be used to operate a stabilization system, such as magnetic bearings, for one or more shafts of the ACM. This provides the advantages of using magnetic bearings without the need for a separate control system, and allows the ACM to achieve higher efficiency. The use of magnetic bearings can allow the ACM to be built without the cooling ducting typically necessary for air bearings, and can further allow the ACM to operate at higher rotational speeds than would be feasible for air bearings.

(11) A geared compressor-turbine connection as described herein allows a preferred rotational speed to be selected for each component. The use of a magnetic gear system (and, in some examples, magnetic bearings) can provide increased reliability and higher ACM efficiency without a large increase in size or weight. While the following description relates to ACM applications, a person skilled in the art would understand the principles discussed herein to apply to, for example, shaft stabilization for a cabin air compressor (CAC) motor with an integrated bearing system, for applications which use less or no bleed air.

(12) FIG. 1 is a schematic depiction of environmental control system (ECS) 10. In the example shown in FIG. 1, ECS 10 is operatively connected to engine 12 and includes air intake 14, primary heat exchanger 16, compressor section 18, secondary heat exchanger 20, turbine section 22, shaft section 24, condenser 26, water collector 28, and air mixer 30. Shaft section 24 can include a compressor shaft and a turbine shaft (such as compressor shaft 220 and turbine shaft 222, both shown in FIG. 4). Compressor section 18 can include compressor flow elements which are disposed about the compressor shaft. Turbine section 22 can include turbine flow elements which are disposed about the turbine shaft. In some examples, the compressor shaft can be axially aligned with the turbine shaft. As described in more detail below, an electrical gearbox containing an axial flux magnetic gear system is used to electromechanically couple elements rotating at different speeds. In the specific embodiments detailed below, the axial flux magnetic gear system can be located between the turbine shaft and the compressor shaft and can electromechanically couple the turbine shaft and the compressor shaft. In some examples, the electrical gearbox can be powered, such that the compressor is able to be driven at a higher speed than would be achieved solely through the rotation of the turbine.

(13) Air intake 14 can include an inlet fan which is fluidly connected to an external environment of the aircraft such that air intake 14 can draw ambient air into ECS 10. Air intake 14 can additionally and/or alternatively include a ram air scoop. Compressor section 18 can include an air compressor which increases the pressure of an airflow as it passes through compressor section 18. Turbine section 22 can include an expansion turbine which removes energy from an airflow as it passes through turbine section 22, decreasing the temperature of the airflow as it expands. Primary heat exchanger 16 and secondary heat exchanger 20 can be air-to-air heat exchangers. Condenser 26 can be fluidly connected to turbine section 22 such that an airflow can exit turbine section 22 and enter condenser 26. Water collector 28 can be fluidly connected to condenser 26 and can be configured to extract water from an airflow. Air mixer 30 can be a chamber fluidly connected to water collector 28, engine 12, and the aircraft cabin such that air mixer 30 can receive conditioned air from water collector 28 and bleed air from engine 12, mix the conditioned air and bleed air to a desired temperature, and deliver it to the aircraft cabin.

(14) ECS 10 can receive a flow of bleed air from a gas turbine engine, and the bleed air is directed into primary heat exchanger 16. In some examples, ECS 10 does not receive bleed air. The bleed air can be cooled with ambient air or ram air from air intake 14. The bleed air which has been

cooled exits primary heat exchanger **16** and is directed to compressor section **18**. Compressor section **18** compresses the cooled air, which increases the cooled air's pressure. The compressed air exits compressor section **18** and is directed to secondary heat exchanger **20** to be further cooled via heat exchange with a flow of ambient air or ram air. The compressed and cooled air exits secondary heat exchanger **20** and enters turbine section **22**. The air expands within turbine section **22**, and the rotation of one or more turbines within turbine section **22** can drive the rotation of a compressor within compressor section **18**. The air exits turbine section **22** and enters condenser **26**. Condenser **26** further cools the flow of air to condense the water vapor contained in the air. The air then enters water collector **28**, which extracts the condensed water from the airflow and thereby dehumidifies the air. The dehumidified air exits water collector **28** and enters air mixer **30** to be mixed with additional, un-conditioned bleed air. This mixing allows the temperature of the resulting conditioned air to be adjusted to a level desirable for an aircraft cabin. The conditioned air is then delivered from air mixer **30** to the aircraft cabin.

(15) As described above, compressor section **18** can include a compressor and turbine section **22** can include at least one turbine, and shaft section **24** can include a compressor shaft and turbine shaft. In this manner, compressor section **18**, turbine section **22**, and shaft section **24** can form a portion of an ACM such as ACM **100** (shown in FIG. 2).

(16) FIG. 2 is a cross-sectional view of ACM **100**. ACM **100** includes compressor **102**, first turbine **104**, second turbine **106**, shaft section **108**, shaft section bearings **110**, ACM housing **112**, first axial diffuser **114**, second axial diffuser **116**, and electrical gearbox **118**. ACM housing **112** includes compressor housing section **120**, first turbine housing section **122**, and second turbine housing portion **124**.

(17) Shaft section **108** can include a compressor shaft and a turbine shaft (such as compressor shaft **220** and turbine shaft **222**, both shown in FIG. 4). Shaft section **108** is oriented along rotational axis X such that compressor **102**, first turbine **104**, and second turbine **106** rotate about rotational axis X. Electrical gearbox **118** is located within shaft section **108** and oriented along rotational axis X. While electrical gearbox **118** is shown in FIG. 2 as being located between compressor **102** and first turbine **104**, electrical gearbox **118** can be located at another suitable location along rotational axis X (such as, for example, between first turbine **104** and second turbine **106**, or between second turbine **106** and compressor **102**) depending upon the design of ACM **100**. Shaft section bearings **110** can support shaft section **108** within sections of ACM housing **112**, such as compressor housing section **120**, first turbine housing section **122**, and/or second turbine housing portion **124**. Shaft section bearings **110** can be magnetic bearings, airfoil bearings, ball bearings, or other suitable bearing systems.

(18) As described above in reference to FIG. 1, the components of ACM **100** work to condition an airflow through an ECS (such as ECS **10**). Compressor **102** compresses the airflow and increases the pressure, and first turbine **104** and second turbine **106** extract energy from the airflow through expansion of the airflow. As described in more detail below, a magnetic gear system within electrical gearbox **118** can drive rotation of compressor **102** through rotation of first turbine **104** and/or second turbine **106** while allowing for differing rotational speeds of compressor **102**, first turbine **104**, and second turbine **106**.

(19) FIGS. 3A-3D illustrate various components of a magnetic gear system within an electrical gearbox (such as electrical gearbox **118** shown in FIG. 2). FIG. 3A is a schematic depiction of high-speed rotor **126**. FIG. 3B is a schematic depiction of low-speed rotor **128**. FIG. 3C is a schematic depiction of stator **130**. FIG. 3D is a schematic depiction of the interaction between high-speed rotor **126** and stator **130** during operation of the magnetic gear system. FIGS. 3A-3D will be discussed in turn below.

(20) While the examples illustrated in FIGS. 3A-3D show an axial flux motor, a person skilled in the art would understand the similar applicability of motors such as radial flux motors. The attractive force between an axial rotor and axial stator can be used to stabilize the rotor position in

combination with a thrust bearing. A radial flux motor including bearings can stabilize a shaft in its radial position (that is, near the center of rotation).

(21) In the example shown in FIG. 3A, high-speed rotor **126** includes magnets **132n**, **132s** distributed circumferentially about rotational axis H. Magnets **132n** are arranged in polarity sections **134n** about high-speed rotor **126**, and magnets **132s** are arranged in polarity sections **134s** about high-speed rotor **126** which are located between polarity sections **134n**. In this manner, polarity sections **134n** and **134s** alternate circumferentially about high-speed rotor **126**. In the example shown in FIG. 3A, high-speed rotor **126** has three pairs of polarity sections **134n**, **134s**. Magnets **132n** are oriented such that the north pole of each magnet is oriented toward a stator of the magnetic gear system (such as stator **130** shown in FIG. 3C). Magnets **132s** are oriented such that the south pole of each magnet is oriented toward a stator of the magnetic gear system (such as stator **130** shown in FIG. 3C). Magnets **132n**, **132s** can be permanent magnets.

(22) In the example shown in FIG. 3B, low-speed rotor **128** includes magnets **136n**, **136s** distributed circumferentially about rotational axis L. Magnets **136n** are arranged in polarity sections **138n** about low-speed rotor **128**, and magnets **136s** are arranged in polarity sections **138s** about low-speed rotor **128** which are located between polarity sections **138n**. Polarity sections **138n** and **138s** thereby alternate circumferentially about low-speed rotor **128**. In the example shown in FIG. 3A, low-speed rotor **128** has nine pairs of polarity sections **138n**, **138s**. Magnets **136n** are oriented such that the north pole of each magnet is oriented toward a stator of the magnetic gear system (such as stator **130** shown in FIG. 3C). Magnets **136s** are oriented such that the south pole of each magnet is oriented toward a stator of the magnetic gear system (such as stator **130** shown in FIG. 3C). Magnets **136n**, **136s** can be permanent magnets.

(23) In the example shown in FIG. 3C, stator **130** includes stator pole sections **140n**, **140s** distributed circumferentially about stator axis S. Stator pole sections **140n**, **140s** can be arranged in pairs such that the polarity of each stator pole section alternates circumferentially about stator **130**. Stator pole sections **140n**, **140s** can be a ferromagnetic material such as steel.

(24) FIG. 3D depicts the interaction between high-speed rotor **126** and stator **130** during operation of electrical gearbox **118**. Magnets **132n** within each polarity section **134n** of high-speed rotor **126** are attracted to stator pole sections **140s** and repelled from stator pole sections **140n**, while magnets **132s** within each polarity section **134s** of high-speed rotor **126** are repelled from stator pole sections **140s** and attracted to stator pole sections **140n**. This alternating cycle causes high-speed rotor **126** to rotate about rotational axis H. Low-speed rotor **128** operates in a substantially similar manner as high-speed rotor **126** with respect to the interaction with stator **130**. The speed of rotation of high-speed rotor **126** and low-speed rotor **128** about their respective rotational axes H, L can depend on the number of pairs of polarity sections present within high-speed rotor **126** and low-speed rotor **128**. High-speed rotor **126** rotates at a higher rotational speed than low-speed rotor **128** due to the lower number of pairs of polarity sections **134n**, **134s** within high-speed rotor **126**. Because each polarity section **134n**, **134s** of high-speed rotor **126** takes up a larger angular percentage of high-speed rotor **126** than the polarity sections **138n**, **138s** of low-speed rotor **128**, there is a greater rotational change between a polarity section **134n** to a polarity section **134s** within high-speed rotor **126**. High-speed rotor **126** thereby travels a greater angular distance for each corresponding stator pole section **140n**, **140s** than low-speed rotor **128**, and accordingly has a higher rotational speed than low-speed rotor **128**.

(25) In some examples, the number of polarity sections **134n**, **134s**, **138n**, **138s** within high-speed rotor **126** and/or low-speed rotor **128** can be varied to select a desired rotational speed of both rotors. As described in more detail below in reference to FIG. 4, the rotor sections can be designed to achieve a desired rotational speed of the compressor. This allows for the gear ratio of the magnetic gear system to be changed through the replacement of one or both rotors. In other examples, magnets **132n**, **132s**, **136n**, **136s** can be electromagnets, and stator pole sections **140n**, **140s** can be permanent magnets. This allows for the polarity of magnets **132n**, **132s**, **136n**, **136s** to

be changed during operation of electrical gearbox **118**, and accordingly allows for the gear ratio of the magnetic gear system to be changed during operation of ECS **10**. Additionally or alternatively, in some examples, electrical gearbox **118** can include a magnetic gear system which has characteristics of both an axial magnetic gear system (such as magnetic gear system **200** shown in FIG. **4**) and a radial magnetic gear system (i.e., a system in which the rotors and stator are arranged coaxially about the rotational axis). Additionally or alternatively, in some examples, stator pole sections **140n**, **140s** can include concentrated windings in a hybrid flux application. These concentrated windings increase the torque of the system by creating additional interactions between the flux generated by each rotor.

(26) FIG. **4** is a schematic cross-sectional depiction of electrical gearbox **118** which includes magnetic gear system **200**, which is oriented along magnetic gear system axis **M**, and gearbox housing **202**. In the example illustrated in FIG. **4**, magnetic gear system **200** includes first rotor **204**, second rotor **206**, stator **208**, backing plates **210**, **212**, winding system **214**, control module **216**, and power source **218**. Compressor shaft **220** and turbine shaft **222** include stabilization systems **224**.

(27) As described above in reference to high-speed rotor **126** and low-speed rotor **128** of FIGS. **3A-3B**, first rotor **204** and second rotor **206** can be made up of magnets which are arranged about the rotational axis of the corresponding rotor, and which are grouped into polarity sections. As described above in reference to stator **130** of FIG. **3C**, stator **208** can be made up of stator pole sections which are arranged in an alternating polarity pattern about stator **208**.

(28) First rotor **204**, second rotor **206**, and stator **208** are each oriented about magnetic gear system axis **M** such that stator **208** is located between first rotor **204** and second rotor **206** along magnetic gear system axis **M** (i.e., such that the stator axis and the rotational axes of the rotors are aligned). Backing plate **210** is located immediately adjacent to first rotor **204** such that first rotor **204** is located between stator **208** and backing plate **210**, and backing plate **212** is located immediately adjacent to second rotor **206** such that second rotor **206** is located between stator **208** and backing plate **212**. Axial air gaps separate first rotor **204**, stator **208**, and second rotor **206** to permit contactless rotation of first rotor **204** and second rotor **206** with respect to stator **208**. Winding system **214** can be, for example, a wire coil which surrounds first rotor **204**, stator **208**, and second rotor **206**.

(29) Control module **216** is a logic-capable device such as a processor or other suitable electronic controller that is electrically connected to at least one of first rotor **204**, stator **208**, second rotor **206**, and winding system **214**. As described in more detail below with respect to FIG. **5**, control module **216** can be configured to control stabilization system **224** in examples where stabilization system **224** includes magnetic bearings. For example, control module **216** can be configured to monitor and adjust the position of the magnetic bearings. Power source **218** can be electrically connected to control module **216**, and additionally or alternatively can be directly electrically connected to at least one of first rotor **204**, stator **208**, second rotor **206**, and winding system **214**. While control module **216** and power source **218** are illustrated in FIG. **4** as being located within electrical gearbox **118**, in some examples control module **216** and/or power source **218** can be externally located to electrical gearbox **118** and can have a wired or wireless connection to electrical gearbox **118** and/or the components within electrical gearbox **118**. While control module **216** and power source **218** are illustrated in FIG. **4** as being discrete and separated components, in some examples control module **216** and power source **218** can be integrated into a single component.

(30) As described above in reference to high-speed rotor **126** and low-speed rotor **128** of FIGS. **3A-3D**, first rotor **204** and second rotor **206** interact with stator **208** such that both first rotor **204** and second rotor **206** rotate about their respective rotational axes. Backing plate **210** is connected to first rotor **204** by suitable means, and accordingly rotates at an equivalent rotational speed to first rotor **204**. Similarly, backing plate **212** is connected to second rotor **206** such that backing plate **212**

and second rotor **206** rotate at an equivalent rotational speed. Backing plate **210** can be connected to turbine shaft **222**, and can in some examples backing plate **210** and turbine shaft **222** can form a monolithic structure. Similarly, backing plate **212** can be connected to compressor shaft **220**, and in some examples backing plate **212** and compressor shaft **220** can form a monolithic structure. First rotor **204** and turbine shaft **222** are accordingly mechanically coupled through the connection with backing plate **210**, and second rotor **206** and compressor shaft **220** are mechanically coupled through the connection with backing plate **212**. Winding system **214** can generate a magnetic field when supplied with an electrical current (for example, from control module **216** and/or power source **218**). In this way, winding system **214** can stabilize the position of first rotor **204** and/or second rotor **206** along each rotor's respective rotational axis. Control module **216** can supply an electrical current to first rotor **204**, stator **208**, second rotor **206**, and/or winding system **214**. This electrical current can be generated by the electromagnetic effects of the magnetic gears' rotation. Stabilization systems **224** can support at least a portion of turbine shaft **222** and/or compressor shaft **220**, and thereby can stabilize the position of turbine shaft **222** and/or compressor shaft **220** along the rotational axis of turbine shaft **222** and/or compressor shaft **220**. Each stabilization system **224** can be, for example, a bearing system (such as bearing system **322** shown in FIG. 5). The bearing system can include magnetic thrust bearings which axially stabilize the shafts, or another suitable bearing type. As described below in reference to FIG. 5, stabilization systems **224** can include magnetic bearings which utilize control module **216**. The magnetic bearings enable very high rotational speeds as compared to air bearings.

(31) In the example depicted in FIG. 4, magnetic gear system **200** is an axial flux magnetic gear system. The rotation of turbine shaft **222** (driven by the rotation of a turbine such as turbines **104**, **106** in FIG. 2) drives the rotation of first rotor **204**. In turn, the rotation of first rotor **204** and second rotor **206** drives the rotation of compressor shaft **220** (which drives the rotation of a compressor such as compressor **102** in FIG. 2). The mechanical coupling of turbine shaft **222** to first rotor **204**, the mechanical coupling of compressor shaft **220** to second rotor **206**, and the electromagnetic coupling of first rotor **204** and second rotor **206** allow for different rotational speeds between compressor shaft **220** and turbine shaft **222** while still allowing the compressor and turbine to be located along a common rotational axis.

(32) In some examples, magnetic gear system **200** can be a hybrid flux magnetic gear system, such as a system which has characteristics of both an axial magnetic gear system and a radial magnetic gear system. This hybrid flux magnetic gear system can, for example, include windings for each stator pole section, adding additional torque to the system. Additionally and/or alternatively, magnetic gear system **200** can provide changeable gear ratios through the use of rotors having differing numbers of polarity sections, as described above in reference to FIGS. 3A-3D. The number of magnets in each of first rotor **204** and second rotor **206** can be selected based on a desired gear ratio of magnetic gear system **200**. While first rotor **204** is illustrated in FIG. 4 as having a lower number of polarity sections than second rotor **206** (and accordingly having a higher rotational speed than second rotor **206**), in some examples first rotor **204** can have a higher number of polarity sections (and accordingly, a lower rotational speed) than second rotor **206**. The number of polarity sections in first rotor **204** and/or second rotor **206** can be varied to ensure an efficient rotational speed for compressor shaft **220** and/or turbine shaft **222**.

(33) FIG. 5 is a schematic cross-sectional depiction of electrical gearbox **300**. Electrical gearbox **300** includes magnetic gear system **302**, which is oriented along magnetic gear system axis M', and gearbox housing **304**. In the example illustrated in FIG. 5, magnetic gear system **302** includes first rotor **306**, second rotor **308**, stator **310**, backing plates **312** and **314**, winding system **316**, control module **318**, power source **320**, and bearing systems **322** for compressor shaft **324** and turbine shaft **326**. Bearing systems **322** can include magnetic bearings **328**, bearing winding systems **330**, and position sensors **332**.

(34) Magnetic gear system **302** can operate in substantially the same manner as magnetic gear

system **200** (described above in reference to FIG. 4). Magnetic bearings **328** serve to stabilize the position of a shaft, such as compressor shaft **324** and/or turbine shaft **326**. Magnetic bearings **328** can include electromagnets which are configured to receive an electric current from power source **320**. This electric current can be altered by control module **318** such that a different amount of current can be applied to individual electromagnets within each magnetic bearing **328**. Bearing winding systems **330** can generate a magnetic field when supplied with an electrical current (for example, from control module **318** and/or power source **320**).

(35) Control module **318** can be configured to control bearing systems **322** via magnetic bearings **328** and position sensors **332**. For example, control module **318** can be configured to monitor the position of a shaft, and can further be configured to adjust the position of the shaft by adjusting current flowing through electromagnets within the magnetic bearings **328**.

(36) Position sensors **332** can detect and monitor the position of a shaft, such as compressor shaft **324** and/or turbine shaft **326**, relative to a magnetic bearing **328** during operation of the air cycle machine. Position sensors **332** can be further configured to communicate with control module **318** to communicate the position of the shaft(s) relative to magnetic bearings **328**. For example, if a shaft has moved closer to or further from a position sensor **332** at a distance that exceeds a threshold distance, position sensor **332** can communicate that data to control module **318**. Control module **318** can then adjust the amount of current directed to one or more electromagnets within magnetic bearings **328** in order to correct the position of the shaft.

(37) An ACM including a magnetic gear system as described herein provides numerous advantages. Incorporating magnetic bearings into an electrical gearbox provides high reliability and higher-precision clearance control without the need for integrating a separate control system. Additionally, decreasing the number of wear surfaces within the shaft bearing system can increase part lifetime and decrease part failure risk. Finally, the use of magnetic bearings does not require the cooling ducts which air bearings require for operation, decreasing system weight.

DISCUSSION OF POSSIBLE EMBODIMENTS

(38) The following are non-exclusive descriptions of possible embodiments of the present invention.

(39) An air cycle machine includes a turbine section, a compressor section, an axial flux magnetic gear system electromechanically coupling the turbine section to the compressor section, and a stabilization system for at least one shaft of the air cycle machine. The axial flux magnetic gear system includes a stator, a first rotor, a second rotor, a winding system, and a control module. The stator includes stator pole sections and is oriented about a stator axis. The first rotor includes a first plurality of magnets arranged radially about a first rotational axis aligned with the stator axis and configured to interact with the stator such that the first rotor rotates at a first rotational speed when the axial flux magnetic gear system is exposed to an electrical current. The second rotor includes a second plurality of magnets arranged radially about a second rotational axis aligned with the first rotational axis and being configured to interact with the stator such that the second rotor rotates at a second rotational speed when the axial flux magnetic gear system is exposed to the electrical current. The winding system stabilizes a position of the first rotor along the first rotational axis and a position of the second rotor along the second rotational axis. The control module is configured to supply the electrical current to at least one of the winding system, the stator, the first rotor, and the second rotor to drive rotation of the first rotor about the first rotational axis and the second rotor about the second rotational axis. The stabilization system stabilizes a position of the at least one shaft of the air cycle machine along a rotational axis of the at least one shaft.

(40) The air cycle machine of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(41) An air cycle machine according to an exemplary embodiment of the present invention, among other possible things, includes a turbine section, a compressor section, an axial flux magnetic gear

system electromechanically coupling the turbine section to the compressor section, and a stabilization system for at least one shaft of the air cycle machine. The axial flux magnetic gear system includes a stator, a first rotor, a second rotor, a winding system, and a control module. The stator includes stator pole sections and is oriented about a stator axis. The first rotor includes a first plurality of magnets arranged radially about a first rotational axis aligned with the stator axis and configured to interact with the stator such that the first rotor rotates at a first rotational speed when the axial flux magnetic gear system is exposed to an electrical current. The second rotor includes a second plurality of magnets arranged radially about a second rotational axis aligned with the first rotational axis and being configured to interact with the stator such that the second rotor rotates at a second rotational speed when the axial flux magnetic gear system is exposed to the electrical current. The winding system stabilizes a position of the first rotor along the first rotational axis and a position of the second rotor along the second rotational axis. The control module is configured to supply the electrical current to at least one of the winding system, the stator, the first rotor, and the second rotor to drive rotation of the first rotor about the first rotational axis and the second rotor about the second rotational axis. The stabilization system stabilizes a position of the at least one shaft of the air cycle machine along a rotational axis of the at least one shaft.

(42) A further embodiment of the foregoing air cycle machine, wherein the stabilization system is a bearing system.

(43) A further embodiment of any of the foregoing air cycle machines, wherein the first rotational speed is greater than the second rotational speed.

(44) A further embodiment of any of the foregoing air cycle machines, wherein the second rotational speed is greater than the first rotational speed.

(45) A further embodiment of any of the foregoing air cycle machines, wherein the first rotor is operatively connected to a turbine of the air cycle machine and the second rotor is operatively connected to a compressor of the air cycle machine such that rotation of the turbine drives rotation of the first rotor and rotation of the second rotor drives rotation of the compressor.

(46) A further embodiment of any of the foregoing air cycle machines, wherein the first rotor comprises a first number of magnets, the second rotor comprises a second number of magnets, and the first number of magnets relative to the second number of magnets is determined by a gear ratio of the axial flux magnetic gear system.

(47) An air cycle machine includes a turbine section, a compressor section, an electrical gearbox, and a stabilization system for at least one of the turbine shaft and the compressor shaft. The turbine section includes a plurality of turbine flow elements disposed about a turbine shaft. The compressor section includes a plurality of compressor flow elements disposed about a compressor shaft axially aligned with the turbine shaft. The electrical gearbox is located between, and electromechanically couples, the turbine shaft and the compressor shaft. The electrical gearbox includes a gearbox housing and an axial flux magnetic gear system comprising disposed within the gearbox housing. The axial flux magnetic gear system includes a stator, a first rotor, a second rotor, a winding system, and a control module. The stator includes a plurality of stator pole sections oriented about a stator axis, and located between the first rotor and the second rotor. The first rotor includes a first plurality of magnets arranged radially about a first rotational axis aligned with the stator axis and configured to interact with the stator such that the first rotor rotates at a first rotational speed when the axial flux magnetic gear system is exposed to an electrical current. The first rotor is mechanically coupled to the turbine shaft. The second rotor includes a second plurality of magnets arranged radially about a second rotational axis aligned with the first rotational axis and being configured to interact with the stator such that the second rotor rotates at a second rotational speed when the axial flux magnetic gear system is exposed to the electrical current. The second rotor is mechanically coupled to the compressor shaft. The winding system stabilizes a position of the first rotor along the first rotational axis and a position of the second rotor along the second rotational axis. The control module is configured to supply the electrical current to at least one of the winding

system, the stator, the first rotor, and the second rotor to drive rotation of the first rotor about the first rotational axis and the second rotor about the second rotational axis. The stabilization system stabilizes a position of the at least one of the turbine shaft and the compressor shaft along a rotational axis of the at least one of the turbine shaft and the compressor shaft.

(48) The air cycle machine of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(49) An air cycle machine according to an exemplary embodiment of the present invention, among other possible things, includes a turbine section, a compressor section, an electrical gearbox, and a stabilization system for at least one of the turbine shaft and the compressor shaft. The turbine section includes a plurality of turbine flow elements disposed about a turbine shaft. The compressor section includes a plurality of compressor flow elements disposed about a compressor shaft axially aligned with the turbine shaft. The electrical gearbox is located between, and electromechanically couples, the turbine shaft and the compressor shaft. The electrical gearbox includes a gearbox housing and an axial flux magnetic gear system comprising disposed within the gearbox housing. The axial flux magnetic gear system includes a stator, a first rotor, a second rotor, a winding system, and a control module. The stator includes a plurality of stator pole sections oriented about a stator axis, and located between the first rotor and the second rotor. The first rotor includes a first plurality of magnets arranged radially about a first rotational axis aligned with the stator axis and configured to interact with the stator such that the first rotor rotates at a first rotational speed when the axial flux magnetic gear system is exposed to an electrical current. The first rotor is mechanically coupled to the turbine shaft. The second rotor includes a second plurality of magnets arranged radially about a second rotational axis aligned with the first rotational axis and being configured to interact with the stator such that the second rotor rotates at a second rotational speed when the axial flux magnetic gear system is exposed to the electrical current. The second rotor is mechanically coupled to the compressor shaft. The winding system stabilizes a position of the first rotor along the first rotational axis and a position of the second rotor along the second rotational axis. The control module is configured to supply the electrical current to at least one of the winding system, the stator, the first rotor, and the second rotor to drive rotation of the first rotor about the first rotational axis and the second rotor about the second rotational axis. The stabilization system stabilizes a position of the at least one of the turbine shaft and the compressor shaft along a rotational axis of the at least one of the turbine shaft and the compressor shaft.

(50) A further embodiment of the foregoing air cycle machine, wherein the stabilization system comprises a bearing system.

(51) A further embodiment of any of the foregoing air cycle machines, wherein the bearing system comprises a plurality of magnetic bearings and a bearing winding system electromechanically coupled to the plurality of magnetic bearings. The plurality of magnetic bearings are oriented about the at least one of the turbine shaft and the compressor shaft, and are operable to control and adjust the position of the at least one of the turbine shaft and the compressor shaft. The control module is configured to supply the electrical current to at least one of the plurality of magnetic bearings and the bearing winding system.

(52) A further embodiment of any of the foregoing air cycle machines, wherein the first rotor is connected to the turbine shaft and the second rotor is connected to the compressor shaft such that rotation of the turbine drives rotation of the first rotor through rotation of the turbine shaft and rotation of the second rotor drives rotation of the compressor through rotation of the compressor shaft.

(53) A further embodiment of any of the foregoing air cycle machines, further comprising a primary heat exchanger, a secondary heat exchanger, a condenser, a water collector, and an air mixer.

(54) A further embodiment of any of the foregoing air cycle machines, further comprising a

stabilization system for at least one of the turbine shaft and the compressor shaft, which stabilizes a position of the at least one of the turbine shaft and the compressor shaft along a rotational axis of the at least one of the turbine shaft and the compressor shaft.

(55) A further embodiment of any of the foregoing air cycle machines, wherein the first rotor comprises a first number of magnets, the second rotor comprises a second number of magnets, and the first number of magnets relative to the second number of magnets is determined by a gear ratio of the axial flux magnetic gear system.

(56) A method of operating an air cycle machine for an aircraft includes generating an electrical current and supplying the electrical current to an axial flux magnetic gear system within an electrical gearbox of the air cycle machine. The axial flux magnetic gear system rotates a compressor shaft of the air cycle machine. A stabilization system stabilizes a position for at least one of a turbine shaft and the compressor shaft of the air cycle machine along a rotational axis of the at least one of the turbine shaft and the compressor shaft. The air cycle machine conditions an airflow for an aircraft cabin of the aircraft.

(57) The method of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(58) A method of operating an air cycle machine for an aircraft according to an exemplary embodiment of the present invention, among other possible things, includes generating an electrical current and supplying the electrical current to an axial flux magnetic gear system within an electrical gearbox of the air cycle machine. The axial flux magnetic gear system rotates a compressor shaft of the air cycle machine. A stabilization system stabilizes a position for at least one of a turbine shaft and the compressor shaft of the air cycle machine along a rotational axis of the at least one of the turbine shaft and the compressor shaft. The air cycle machine conditions an airflow for an aircraft cabin of the aircraft.

(59) A further embodiment of the foregoing method, wherein supplying the electrical current to the axial flux magnetic gear system within the electrical gearbox of the air cycle machine comprises supplying the electrical current to at least one of a winding system of the axial flux magnetic gear system, a stator of the axial flux magnetic gear system, a first rotor of the axial flux magnetic gear system, and a second rotor of the axial flux magnetic gear system. The first rotor is configured to interact with a stator of the axial flux magnetic gear system such that the first rotor rotates about a magnetic gear system axis at a first rotational speed when the axial flux magnetic gear system is exposed to the electrical current. The stator comprises a plurality of stator pole sections and is oriented along the magnetic gear system axis. The second rotor is configured to interact with the stator such that the second rotor rotates about the magnetic gear system axis at a second rotational speed when the axial flux magnetic gear system is exposed to the electrical current. The winding system stabilizes a position of the first rotor and a position of the second rotor along the magnetic gear system axis. Rotating the compressor shaft of the air cycle machine using the axial flux magnetic gear system comprises rotating a turbine shaft through operation of a turbine of the air cycle machine, rotating the first rotor about the magnetic gear system axis through rotation of the turbine shaft, rotating the second rotor about the magnetic gear system axis, and rotating the compressor shaft through rotation of the second rotor. The first rotor is mechanically coupled to the turbine shaft. The second rotor is mechanically coupled to the compressor shaft.

(60) A further embodiment of any of the foregoing methods, wherein conditioning the airflow for the aircraft cabin of the aircraft using the air cycle machine comprises receiving a flow of air which is bleed air from an engine of the aircraft, directing the flow of air through a primary heat exchanger, compressing the flow of air with a compressor of the air cycle machine which is connected to the compressor shaft, directing the flow of air through a turbine of the air cycle machine, and delivering the flow of air to the aircraft cabin.

(61) A further embodiment of any of the foregoing methods, wherein conditioning the airflow for

the aircraft cabin of the aircraft using the air cycle machine further comprises directing the flow of air through a secondary heat exchanger, condensing water vapor in the flow of air using a condenser of the air cycle machine, collecting water from the flow of air using a water collector of the air cycle machine, and mixing the flow of air with bleed air from the engine before delivering the flow of air to the aircraft cabin.

(62) A further embodiment of any of the foregoing methods, wherein the stabilization system comprises a bearing system. The bearing system comprises a plurality of magnetic bearings and a bearing winding system electromechanically coupled to the plurality of magnetic bearings. The plurality of magnetic bearings are oriented about the at least one of the turbine shaft and the compressor shaft, and are operable to control and adjust the position of the at least one of the turbine shaft and the compressor shaft. The control module supplies the electrical current to at least one of the plurality of magnetic bearings and the bearing winding system. The bearing system adjusts the position of the at least one of the turbine shaft and the compressor shaft.

(63) While the invention has been described with reference to an exemplary embodiment(s), it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment(s) disclosed, but that the invention will include all embodiments falling within the scope of the appended claims.

Claims

1. An air cycle machine comprising: a turbine section; a compressor section; an axial flux magnetic gear system electromechanically coupling the turbine section to the compressor section, the axial flux magnetic gear system comprising: a stator comprising a plurality of stator pole sections and oriented about a magnetic gear system axis such that the plurality of stator pole sections are distributed circumferentially about the magnetic gear system axis; a first rotor comprising a first plurality of magnets distributed circumferentially about the magnetic gear system axis and configured to interact with the stator such that the first rotor rotates at a first rotational speed when the axial flux magnetic gear system is exposed to an electrical current; a second rotor comprising a second plurality of magnets distributed circumferentially about the magnetic gear system axis and configured to interact with the stator such that the second rotor rotates at a second rotational speed when the axial flux magnetic gear system is exposed to the electrical current; a winding system operable to stabilize a position of the first rotor and a position of the second rotor along the magnetic gear system axis; and a control module capable of supplying the electrical current to at least one of the winding system, the stator, the first rotor, and the second rotor, thereby driving rotation of the first rotor at the first rotational speed, and of the second rotor at the second rotational speed, about the magnetic gear system axis; and a stabilization system for at least one shaft of the air cycle machine which stabilizes a position of the at least one shaft of the air cycle machine along a rotational axis of the at least one shaft.

2. The air cycle machine of claim 1, wherein the stabilization system is a bearing system.

3. The air cycle machine of claim 1, wherein: the bearing system comprises: a plurality of magnetic bearings oriented about the at least one shaft and operable to control and adjust the position of the at least one shaft; and a bearing winding system electromechanically coupled to the plurality of magnetic bearings; and the control module is configured to supply the electrical current to at least one of the plurality of magnetic bearings and the bearing winding system.

4. The air cycle machine of claim 1, wherein the first rotational speed is greater than the second rotational speed.

5. The air cycle machine of claim 1, wherein the second rotational speed is greater than the first

rotational speed.

6. The air cycle machine of claim 1, wherein the first rotor is operatively connected to a turbine of the air cycle machine and the second rotor is operatively connected to a compressor of the air cycle machine such that rotation of the turbine drives rotation of the first rotor and rotation of the second rotor drives rotation of the compressor.
7. The air cycle machine of claim 1, wherein: the first rotor comprises a first number of magnets; the second rotor comprises a second number of magnets; and the first number of magnets relative to the second number of magnets is determined by a gear ratio of the axial flux magnetic gear system.
8. An air cycle machine comprising: a turbine section comprising a plurality of turbine flow elements disposed about a turbine shaft; a compressor section comprising a plurality of compressor flow elements disposed about a compressor shaft axially aligned with the turbine shaft; an electrical gearbox located between and electromechanically coupling the turbine shaft and the compressor shaft, the electrical gearbox comprising: a gearbox housing; and an axial flux magnetic gear system comprising disposed within the gearbox housing and comprising: a stator comprising a plurality of stator pole sections oriented about a magnetic gear system axis such that the plurality of stator pole sections are distributed circumferentially about the magnetic gear system axis, and located between a first rotor and second rotor; the first rotor, comprising a first plurality of magnets distributed circumferentially about the magnetic gear system axis and configured to interact with the stator such that the first rotor rotates at a first rotational speed when the axial flux magnetic gear system is exposed to an electrical current, wherein the first rotor is mechanically coupled to the turbine shaft; the second rotor, comprising a second plurality of magnets distributed circumferentially about the magnetic gear system axis and configured to interact with the stator such that the second rotor rotates at a second rotational speed when the axial flux magnetic gear system is exposed to the electrical current, wherein the second rotor is mechanically coupled to the compressor shaft; a winding system operable to stabilize a position of the first rotor and a position of the second rotor along the magnetic gear system axis; and a control module capable of supplying the electrical current to at least one of the winding system, the stator, the first rotor, and the second rotor, thereby driving rotation of the first rotor at the first rotational speed, and of the second rotor at the second rotational speed, about the magnetic gear system axis; and a stabilization system for at least one of the turbine shaft and the compressor shaft, which stabilizes a position of the at least one of the turbine shaft and the compressor shaft along a rotational axis of the at least one of the turbine shaft and the compressor shaft.
9. The air cycle machine of claim 8, wherein the stabilization system comprises a bearing system.
10. The air cycle machine of claim 9, wherein: the bearing system comprises: a plurality of magnetic bearings oriented about the at least one of the turbine shaft and the compressor shaft and operable to control and adjust the position of the at least one of the turbine shaft and the compressor shaft; and a bearing winding system electromechanically coupled to the plurality of magnetic bearings; and the control module is configured to supply the electrical current to at least one of the plurality of magnetic bearings and the bearing winding system.
11. The air cycle machine of claim 8, wherein the first rotor is connected to the turbine shaft and the second rotor is connected to the compressor shaft such that rotation of the turbine drives rotation of the first rotor through rotation of the turbine shaft and rotation of the second rotor drives rotation of the compressor through rotation of the compressor shaft.
12. The air cycle machine of claim 8, further comprising: a primary heat exchanger; a secondary heat exchanger; a condenser; a water collector; and an air mixer.
13. The air cycle machine of claim 8, wherein: the first rotor comprises a first number of magnets; the second rotor comprises a second number of magnets; and the first number of magnets relative to the second number of magnets is determined by a gear ratio of the axial flux magnetic gear system.

